ALICE L WALTON SCHOOL OF MEDICINE

Gastric Secretion and the Parietal Cell

Last updated July 9, 2026 · 17 min read

CC BY-SA 4.0

Authors: Ryan Downey, Lance Bridges
Editors: Jordan Finn, Russell Johnson
Contributors: Ian Murray, Eleanor Schrems, Kevin Kunkler

Listen to this Brick
0:00 / 34:12 1x

Learning Objectives (5)

After completing this brick, you will be able to:

1. Identify the major gastric secretory cell types, their products, and their regional distribution within the gastric mucosa, and describe the motility pattern of each gastric region.
2. Explain the parietal cell mechanism of HCl secretion, including the roles of H⁺/K⁺[illegible] transport.
3. Analyze how gastrin, histamine, acetylcholine, and somatostatin integrate at the parietal cell to regulate acid secretion, including the role of GRP as the vagal mediator of gastrin release from G cells.
4. Explain the mechanism of pepsinogen secretion and activation, and predict the consequences of achlorhydria on the initiation of protein digestion.
5. Predict how failure of specific gastric mucosal defense mechanisms leads to epithelial injury, based on their normal protective function.

CASE CONNECTION

JM, a 35-year-old male, presents at the outpatient clinic with a 6-week history of persistent epigastric pain and occasional nausea. He denies vomiting or weight loss. On upper endoscopy, patchy mucosal erythema is noted in the gastric body and antral regions.

What symptoms might arise if gastric acid secretion is impaired? What risks are present if gastric mucosal defenses fail? Consider your answers as you read, and we'll revisit JM at the end of the brick.

GO TO CONCLUSION ↓

Gastric Secretory Cell Types, Products, and Regional Distribution (LO1)

Cell Types and Products

The gastric mucosa contains six major secretory cell populations. Parietal cells (oxyntic cells) secrete hydrochloric acid (HCl) and intrinsic factor (IF): HCl creates the acidic environment required for protein denaturation and pepsin activation; IF is required for vitamin B12 absorption in the terminal ileum. Chief cells (zymogenic cells) secrete pepsinogen and gastric lipase; pepsinogen is the inactive precursor to pepsin and initiates protein digestion following acid-dependent activation, while gastric lipase contributes to intragastric fat digestion.

G cells are enteroendocrine cells that secrete gastrin, which potently stimulates acid secretion. Enterochromaffin-like (ECL) cells secrete histamine, the dominant paracrine stimulant of parietal cell acid secretion. D cells secrete somatostatin, a local inhibitor of both gastrin release and acid secretion. Mucous neck cells and surface mucous cells secrete mucus and bicarbonate, forming the adherent gel layer that protects the epithelium from acid-mediated injury.

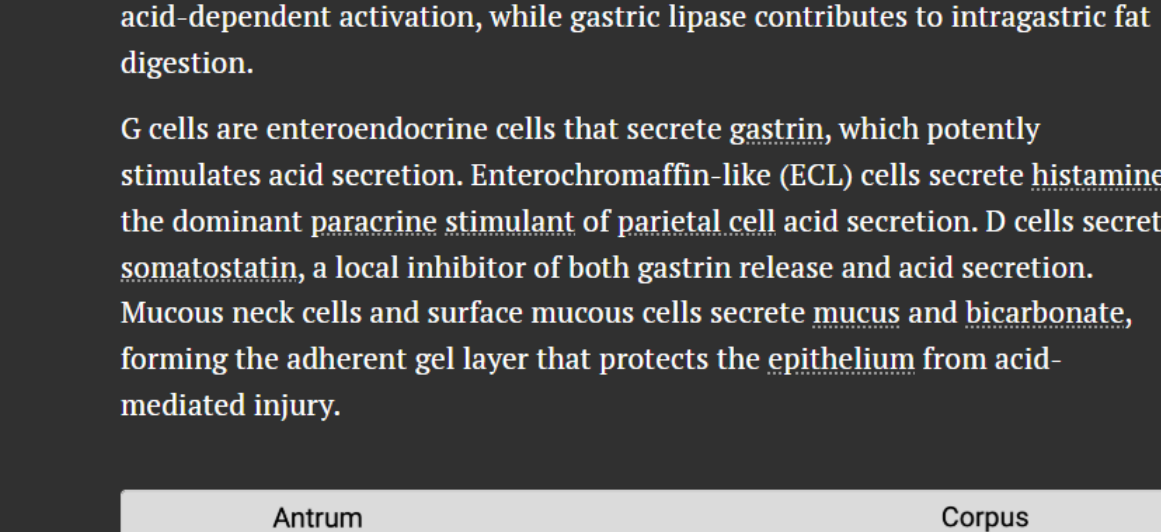

Figure 1: Di[illegible]stric glands.

Table 1: Cell types and functions

Cell Type	Secretory Product(s)	Primary Function	Primary Location
Parietal cell	HCl, intrinsic factor	Acid secretion, B12 absorption	Body/fundus
Chief cell	Pepsinogen	Protein digestion (after activation)	Body/fundus
Enterochromaffin-like (ECL) cell	Histamine	Dominant paracrine stimulant of parietal cell acid secretion	Body/fundus
G cell	Gastrin	Stimulates acid secretion	Antrum
D cell	Somatostatin	Inhibits acid and gastrin release	Antrum & body
Mucous cells	Mucus, bicarbonate	Mucosal protection	Throughout

Regional Distribution

The combined secretory products of the gastric mucosa — acid, pepsin, mucus, and gastric lipase — mix with ingested food to form chyme, the semi-liquid material that passes from the stomach into the duodenum.

The stomach is divided into three regions: the cardia, the body (corpus) and fundus, and the antrum/pylorus. The body and fundus contain oxyntic glands densely packed with [illegible]s — the primary site of acid, pepsinogen, gastric lipase, and histamine secretion; D cells here provide local inhibitory feedback. The antrum and pylorus contain pyloric glands rich in G cells and D cells: G cells sense luminal peptides and amino acids and release gastrin into the bloodstream to stimulate parietal cells in the body, while D cells in both regions modulate the response. The cardia contains primarily mucous cells with minimal secretory activity.

Match each gastric cell type to its secretory product and primary gastric region.

ACTIVE LEARNING QUESTION

Match each gastric cell type to its secretory product and primary gastric region.

HCl & IF, body/fund[illegible]
pepsinogen, body/fundus
gastrin, antrum

Parietal cell	
Chief cell	
G cell	
D cell	
Mucous cells	

Explanation (requires correct answer)

Parietal Cell Mechanism of HCl Secretion (LO2)

The parietal cell generates luminal H⁺ concentrations more than one million times greater than plasma (lumen pH ~1–2 vs. plasma pH ~7.4) through the coordinated action of H⁺/K⁺-ATPase, carbonic anhydrase, and a basolateral chloride transport system.

H⁺/K⁺–ATPase Function

H⁺/K⁺-ATPase, the apical membrane proton pump, uses ATP hydrolysis to exchange intracellular H⁺ for luminal K⁺ in a 1:1 ratio, generating the pH ~1–2 environment of gastr[illegible]Pase resides in intracellular tubulovesicular membranes; upon stimulation, these vesicles fuse with the apical membrane, increasing pump density and acid output.

Carbonic Anhydrase

H⁺ supply is maintained by the intracellular hydration of CO₂, catalyzed by carbonic anhydrase: $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$. The H⁺ is secreted into the lumen via H⁺/K⁺-ATPase; HCO_3^- is exported across the basolateral membrane into the bloodstream in exchange for Cl⁻ via a Cl⁻/HCO_3^- antiporter. This basolateral HCO_3^- efflux produces the transient rise in blood and urine pH known as the alkaline tide.

Chloride Transport

Cl⁻ enters via the basolateral Cl⁻/HCO_3^- exchanger and exits through apical Cl⁻ channels, combining with secreted H⁺ in the lumen to form HCl. K⁺ pumped into the cell by H⁺/K⁺-ATPase is recycled back into the lumen through apical K⁺ channels, maintaining luminal K⁺ availability as substrate for continued pump activity.

CLINICAL CORRELATION

Acid-Suppressing Drug Classes: Three classes act at distinct points in the acid secretion pathway. *Antacids* (e.g., calcium carbonate, aluminum hydroxide) act in the gastric lumen, directly neutralizing secreted HCl; they provide rapid but short-lived relief. *H₂ receptor antagonists* (e.g., famotidine) block histamine's H₂ receptor on parietal cells, reducing cAMP-driven acid secretion. *Proton pump inhibitors* (e.g., omeprazole) ir[illegible]iding the most sustained acid suppression. This mechanistic hierarchy explains their differing onset, duration, and clinical indications.

ACTIVE LEARNING QUESTION

A patient is prescribed a PPI that irreversibly inhibits H⁺/K⁺-ATPase. Which intracellular process is most directly impaired in parietal cells?

A. ○ Carbonic anhydrase activity
B. ○ Apical H⁺ secretion
C. ○ Basolateral Cl⁻ entry
D. ○ Basolateral HCO_3^- export
E. ○ Apical K⁺ recycling

Explanation (requires correct answer)

Regulation of Acid Secretion at the Parietal Cell (LO3)

Acid secretion is regu[illegible]astrin, histamine, and acetylcholine—and one major inhibitory signal, somatostatin. These pathways converge on the parietal cell and interact to modulate acid output in response to meals and neural input.

Regulatory Signal Pathways

Gastrin, released from antral G cells in response to luminal peptides and amino acids, acts via the bloodstream on CCK-B receptors on parietal cells and ECL cells. At parietal cells, gastrin directly stimulates acid secretion; at ECL cells, it triggers histamine release,amplifying the response. Histamine acts on H₂ receptors, a GPCR, on parietal cells, activating adenylyl cyclase → cAMP → PKA, ultimately driving H⁺/K⁺-ATPase translocation to the apical membrane. Histamine is the dominant stimulatory route to the parietal cell: it provides the central potentiating drive that amplifies the other signals, and much of gastrin's effect on acid secretion is itself mediated through ECL histamine release. This is why H₂ receptor antagonists sharply reduce both basal and meal-stimulated acid output.

Acetylcholine (ACh), released from ENS motor neurons activated by vagal input, binds M₃ muscarinic receptors on parietal cells, activating PLC → IP₃/DAG → elevated Ca²⁺ and PKC. ACh also stimulates ECL cells to release histamine and inhibits somatostatin release from D cells, removing a key brake on both G cells and parietal cells, creating multilevel amplification of the acid response. Somatostatin, released from D cells in response to low luminal pH, suppresses gastrin release from G cells and directly inhibits parietal cell activity via Gi-coupled receptors that reduce cAMP. These cascades are summarized in the figure and table below.

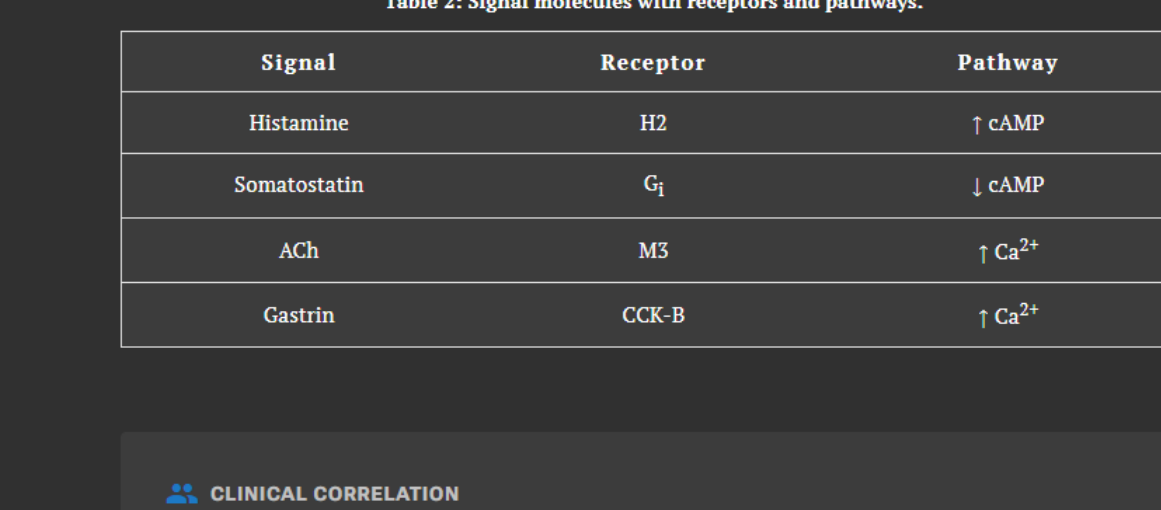

Figure 2: HCl release pathway in GI parietal cells.

Table 2: Signal molecules with receptors and pathways.

Signal	Receptor	Pathway
Histamine	H2	↑ cAMP
Somatostatin	G_i	↓ cAMP
ACh	M3	↑ Ca²⁺
Gastrin	CCK-B	↑ Ca²⁺

CLINICAL CORRELATION

H₂ Receptor Antagonists: Because the histamine → H₂ receptor → cAMP pathway is the domi[illegible]H₂ receptor antagonists (e.g., famotidine) effectively reduce basal and meal-stimulated acid secretion. Note that they do not block gastrin or ACh pathways directly, which limits their efficacy compared to PPIs.

GRP and Vagal Influence

During the cephalic phase, ENS neurons in the antrum, activated by vagal input, release gastrin-releasing peptide (GRP), which binds GRP receptors on G cells and triggers gastrin secretion. This GRP–gastrin axis is the primary mechanism by which the CNS initiates gastric acid secretion in anticipation of food, prior to any luminal stimulus. Because GRP signals through a non-muscarinic pathway, this response is not blocked by atropine.

ACTIVE LEARNING QUESTION

A researcher blocks vagal GRP signaling in an animal model. Which effect on gastric acid secretion is most expected during the cephalic phase?

A. ○ Increased direct parietal ACh stimulation
B. ○ Increased ECL histamine release
C. ○ Reduced gastrin-stimulated acid secretion
D. ○ Increased somatostatin inhibition
E. ○ Decreased intrinsic factor secretion

Explanation (requires correct answer)

Pepsinogen Secretion, Activation, and Consequences of Achlorhydria (LO4)

Protein digestion in the stomach depends on the coordinated action of chief cells and parietal cells. Chief cells produce and secrete pepsinogen, a zymogen (an inactive precursor of an enzyme). Its conversion to active pepsin requires the acidic luminal environment generated by parietal cells, confining proteolytic activity to the gastric lumen and protecting the epithelium from autodigestion.

Pepsinogen Release

Chief cells, located predominantly in the gastric body and fundus, store pepsinogen and gastric lipase in apical secretory granules. Gastric lipase, an acid-stable lipase active at pH 3–6, initiates intragastric fat digestion by cleaving triglycerides into fatty acids and diglycerides, accounting for roughly 10–30% of total fat digestion. Pepsinogen secretion is stimulated by ACh (via vagal and enteric neural input), gastrin, and secretin; stimulation triggers exocytosis into the gastric lumen.

Activation

At luminal pH below 5 (optimally pH 1.8–3.5), H⁺ cleaves an inhibitory peptide from pepsinogen, exposing the active site and generating pepsin. Pepsin then cleaves additional pepsinogen molecules autocatalytically, rapidly amplifying proteolytic activity. Active pepsin cleaves peptide bonds adjacent to aromatic amino acids. Pepsin activity is irreversibly lost at pH above 7, which occurs as chyme enters the alkaline duodenum.

CLINICAL REASONING

Pepsin has relatively broad enzymatic specificity, preferentially cleaving peptide bonds adjacent to hydrophobic, particularly aromatic amino acids (e.g., phenylalanine, tryptophan, tyrosine). This broad specificity allows efficient digestion of proteins from diverse dietary sources, including both animal and plant proteins. In contrast, individual pancreatic proteases exhibit more specific cleavage preferences; however, collectively they exhibit very broad proteolytic activity.

Consequences of Achlorhydria

Achlorhydria — absence of gastric acid secretion — prevents pepsinogen activation, eliminating the gastric phase of protein digestion. The small intestine can partially compensate through pancreatic proteases (trypsin, chymotrypsin, elastase), though this compensation may be incomplete. Achlorhydria also impairs absorption of nutrients requiring an acidic luminal environment: the acidic environment generated by HCl solubilizes dietary iron and facilitates its red[illegible] achlorhydria impairs non-heme iron absorption; when IF secretion is concurrently reduced, vitamin B12 absorption is also impaired.

CLINICAL CORRELATION

Autoimmune Gastritis: In autoimmune gastritis, autoantibodies destroy parietal cells, resulting in combined achlorhydria and loss of intrinsic factor secretion. The loss of IF causes vitamin B12 malabsorption (pernicious anemia), while impaired acid secretion reduces non-heme iron absorption. This is a clinically important example of how parietal cell dysfunction affects multiple absorptive processes simultaneously.

What triggers pepsinogen activation, and what happens to pepsin activity when chyme enters the duodenum?

Gastric Mucosal Defense and Pathways to Epithelial Injury (LO5)

The gastric mucosa is continuously exposed to concentrated HCl, pepsin, and ingested irritants; its integrity depends on a multilayered defense system. When any component fails, the result can range from superficial erosions to ulceration.

Protective Mechanisms

The mucus-bicarbonate barrier is the primary defense layer: surface mucous cells secrete an adherent mucus gel that traps secreted HCO_3^-, establishing a pH gradient from ~2 at the lumen to ~7 at the epithelial surface. Prostaglandins (PGE_2 and PGI_2) stimulate mucus and bicarbonate secretion and maintain mucosal blood flow. Tight junctions between epithelial cells prevent H⁺ back-diffusion. Epithelial restitution—migration of surface cells to reseal breaches within minutes—provides a third layer. Mucosal blood flow delivers oxygen and nutrients while clearing any H⁺ that penetrates the epithelium. These mechanisms are synergistic; failure of any single layer increases vulnerability to the others.

Pathways to Injury

NSAIDs inhibit COX-1 enzymes, reducing prostaglandin synthesis and simultaneously decreasing mucus and bicarbonate secretion, reducing mucosal blood flow, and impairing restitution. *H. pylori* disrupts the mucus layer via urease-produced ammonia, damages tight junctions through inflammation, and creates a hypersecretory state by stimulating gastrin release and suppressing somatos[illegible]olubilize the mucus layer and disrupt tight junctions.

Once H⁺ penetrates the epithelial barrier, mast cell degranulation releases histamine, which paradoxically amplifies the injury by further stimulating acid secretion while increasing vascular permeability; mucosal edema and hemorrhage follow, and sustained injury leads to ulceration.

CLINICAL CORRELATION

COX Isoforms and GI Toxicity: COX-1 is constitutively expressed in gastric mucosa and produces the prostaglandins that drive mucosal protection; COX-2 is induced at inflammatory sites. Non-selective NSAIDs inhibit both isoforms, simultaneously reducing inflammation and impairing mucosal defense. COX-2 selective inhibitors (coxibs) preserve gastric prostaglandin synthesis, reducing GI toxicity — though they carry increased cardiovascular risk. This isoform distinction is central to understanding NSAID pharmacology and the rationale for co-prescribing PPIs in patients requiring long-term NSAID therapy.

CLINICAL CORRELATION

NSAID-Induced Gastropathy and *H. pylori*: NSAIDs and *H. pylori* infection are the two leading causes of peptic ulcer disease, acting through complementary mechanisms of mucosal defense failure. Both conditions impair prostaglandin-mediated protection; H. pylori adds the additional insult of hypersecretion. Restoration of prostaglandin signaling (e.g., synthetic prostaglandin analogues) and H. pylori eradication are key therapeutic targets.

ACTIVE LEARNING QUESTION

A patient on chronic NSAID therapy develops a gastric ulcer. Which defense mechanism is most directly impaired by NSAIDs?

A. ○ Prostaglandin-stimulated mucus/HCO_3^-
B. ○ Tight junction integrity
C. ○ H⁺/K⁺-ATPase activity
D. ○ Pepsinogen secretion
E. ○ Epithelial restitution

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JM, what symptoms might arise if gastric acid secretion is impaired? What risks are present if gastric mucosal defenses fail? JM's patchy mucosal erythema in the gastric body and antrum suggests inflammation affecting both parietal/chief ce[illegible]ired acid secretion in the body could re[illegible]g initial protein digestion and impairing iron and B12 absorption. If antral D cell function is disrupted, loss of somatostatin-mediated inhibition could lead to gastrin excess and paradoxical hypersecretion.

The erythema also signals mucosal defense compromise: disrupted mucus-bicarbonate barriers and tight junctions allow H⁺ back-diffusion, driving the epithelial injury and pain JM experiences. Understanding these mechanisms points toward targeted interventions—PPIs to reduce acid load, prostaglandin analogs to restore mucosal defense, and evaluation for H. pylori as an underlying cause of JM's persistent symptoms.

Summary

- The gastric mucosa contains six major secretory cell types: parietal cells (HCl, intrinsic factor), chief cells (pepsinogen), and enterochromaffin-like (ECL) cells (histamine) in the body/fundus, G cells (gastrin) and D cells (somatostatin) in the antrum, and mucous cells throughout. Each region also has distinct motility patterns—proximal accommodation and distal grinding/propulsion.
- Parietal cells secrete HCl via apical H⁺/K⁺-ATPase, fueled by carbonic anhydrase-generated H⁺. Cl⁻ enters basolaterally via Cl⁻/HCO_3^- exchange and exits apically to combine with H⁺, forming HCl. The resulting HCO_3^- efflux produces the postprandial alkaline tide.
- Acid secretion is stimulated by gastrin (CCK-B receptor, via PLC/Ca²⁺), histamine (H₂ receptor, via cAMP/PKA), and acetylcholine (M₃ receptor, via IP₃/Ca²⁺). Somatostatin inhibits both G cells and parietal cells. GRP—not ACh—is the vaga[illegible]e from antral G cells during the cephalic phase.
- Pepsinogen is secreted by chief cells and activated by luminal H⁺ (pH <5) and autocatalysis. Active pepsin initiates protein digestion and is inactivated at pH >7 in the duodenum. Achlorhydria prevents pepsinogen activation, impairing gastric protein digestion and reducing absorption of acid-dependent nutrients.
- Gastric mucosal defense relies on the mucus-bicarbonate barrier, tight junctions, epithelial restitution, and mucosal blood flow—all supported by prostaglandins. NSAIDs impair prostaglandin synthesis, simultaneously weakening multiple defense layers. H. pylori disrupts mucus, alters the gastrin/somatostatin balance, and promotes inflammation, leading to ulceration.

Review Questions

1. Which gastric cell is found mainly in the antrum and secretes gastrin?

A. ○ Parietal cell
B. ○ Chief cell
C. ○ G cell
D. ○ D cell
E. ○ Mucous cell

Explanation (requires correct answer)

2. What process does the H+/K+-ATPase directly perform in parietal cells?

A. ○ Carbonic anhydrase activity
B. ○ Apical H+ secretion
C. ○ Basolateral Cl- entry
D. ○ Basolateral HCO3- export
E. ○ Apical K+ recycling

Explanation (requires correct answer)

3. Blocking vagal GRP signaling during the cephalic phase would most directly prevent which normal response?

A. ○ Increased parietal ACh stimulation
B. ○ Increased ECL histamine release
C. ○ Reduced gastrin-stimulated acid secretion
D. ○ Increased somatostatin release from D cells
E. ○ Decrease[illegible]

Explanation (requires correct answer)

4. What happens to protein digestion if gastric acid secretion fails?

A. ○ Pepsinogen is not activated
B. ○ Pepsinogen secretion is decreased
C. ○ Intrinsic factor secretion decreases
D. ○ Pancreatic proteases decrease
E. ○ Somatostatin inhibition increases

Explanation (requires correct answer)

5. NSAIDs directly impair which gastric mucosal defense?

A. ○ Prostaglandin-stimulated mucus/HCO3-
B. ○ Tight junction integrity
C. ○ H+/K+-ATPase activity
D. ○ Pepsinog[illegible]
E. ○ Epithelial restitution

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Identify the major gastric secretory cell types, their products, and their regional distribution within the gastric mucosa, and describe the motility pattern of each gastric region.
02 ☐ Explain the parietal cell mechanism of HCl secretion, including the roles of H⁺/K⁺-ATPase, carbonic anhydrase, and Cl⁻ transport.
03 ☐ Analyze how gastrin, histamine, acetylcholine, and somatostatin integrate at the parietal cell to regulate acid secretion, including the role of GRP as the vagal mediator of gastrin release from G cells.
04 ☐ Explain the mechanism of pepsinogen secretion and activation, and predict the consequences of achlorhydria on the initiation of protein[illegible]

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Identify the major gastric secretory cell types, their products, and their regional distribution within the gastric mucosa, and describe the motility pattern of each gastric region.
02 ☐ Explain the parietal cell mechanism of HCl secretion, including the roles of H⁺/K⁺-ATPase, carbonic anhydrase, and Cl⁻ transport.
03 ☐ Analyze how gastrin, histamine, acetylcholine, and somatostatin integrate at the parietal cell to regulate acid secretion, including the role of GRP as the vagal mediator of gastrin release from G cells.
04 ☐ Explain the mechanism of pepsinogen secretion and activation, and predict the consequences of achlorhydria on the initiation of protein digestion.
05 ☐ Predict how failure of specific gastric mucosal defense mechanisms leads to epithelial injury, based on their normal protective function.

Objective Confidence: 0/5